

Ideation



Design Constraints

- 12x12x18 inches
- Must be able to launch a ping ball of ~1.57in
- Ball reservoir must hold at least 5 balls at once

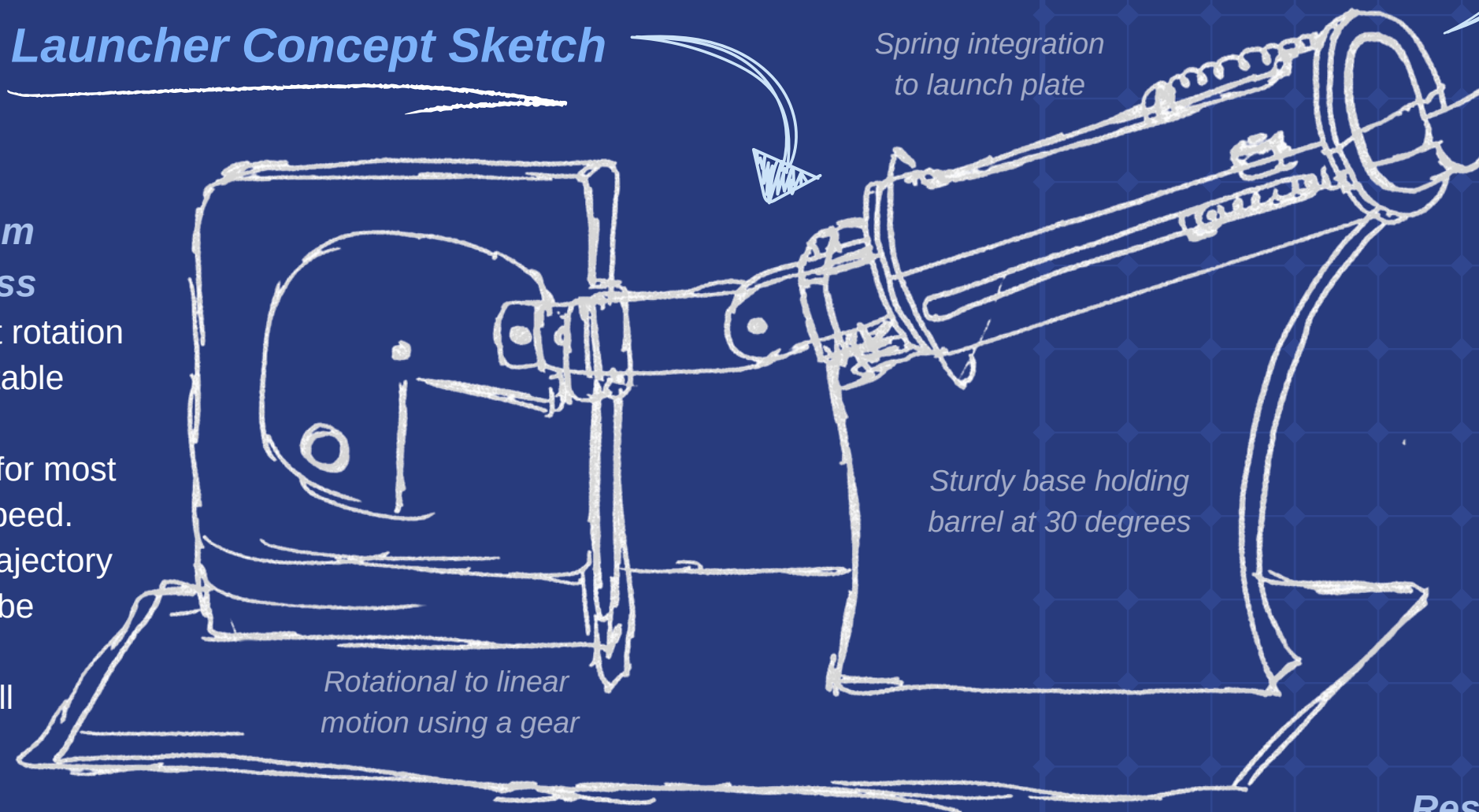
Driving Questions

- What number of springs will reliably launch ping pong balls 8 feet using our designed mechanism?
- What is the allowable range of launch angles that still enables the ball to land in the target zone at an 8-ft distance?

1) Launcher Concept Sketch

Launch mechanism metrics for success

- Efficiently convert rotation to smooth, repeatable linear release.
- Keep friction low for most efficient launch speed.
- Maintain stable trajectory with consistent tube guidance.
- Ensure proper ball clearance
- Strong hardware integration to avoid weak PLA pieces breaking from spring force



2) Reservoir Concept Sketch

Ball ramp
& reservoir



Reservoir Concept V.1

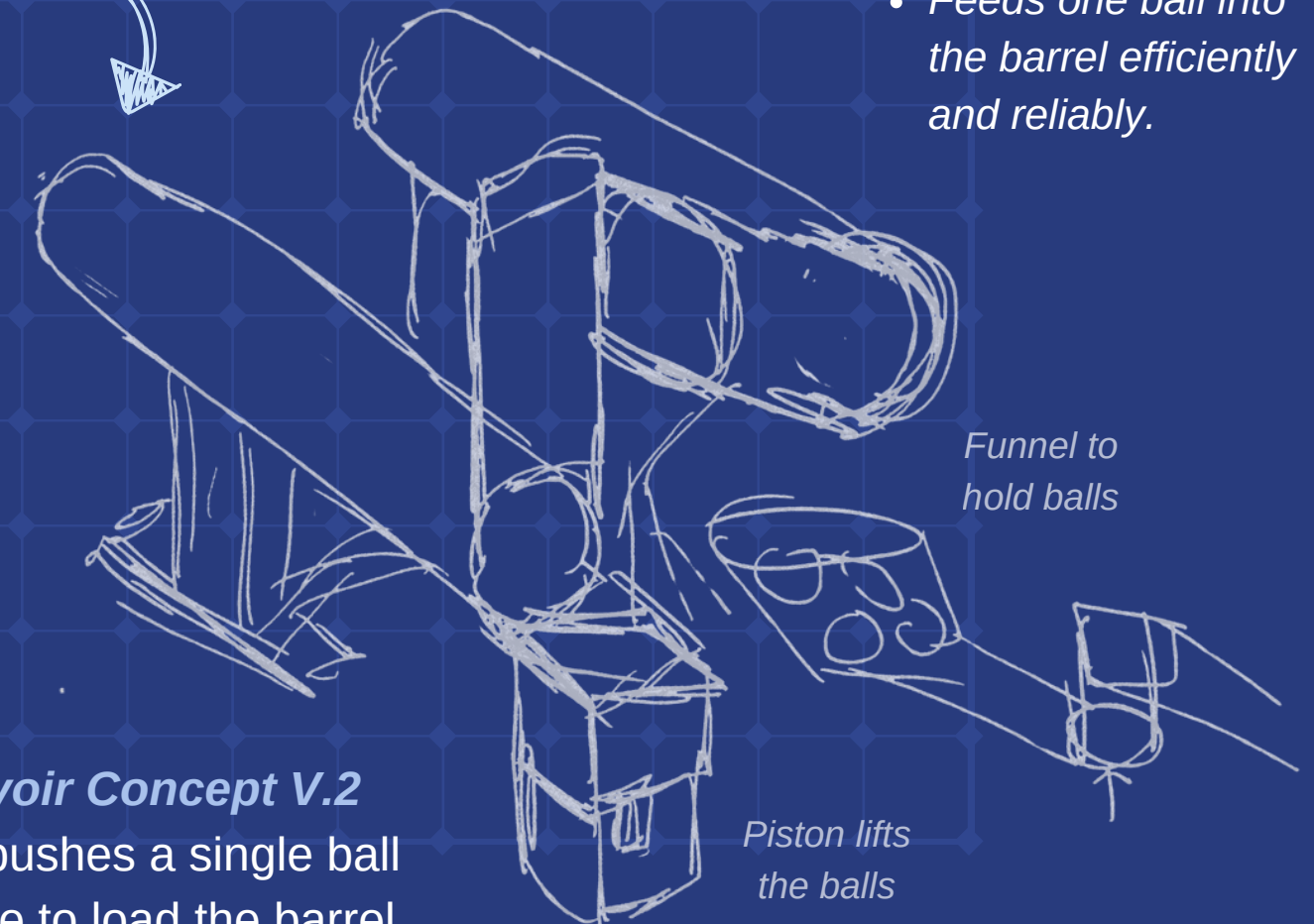
Push one ball over a lip at a time to load it into the barrel for launch

Ball reservoir mechanism metrics for success

- Holds at least 5 balls before launch.
- Secures non-launched balls so they don't fall out or interfere with firing.
- Feeds one ball into the barrel efficiently and reliably.

Reservoir Concept V.2

Hopper pushes a single ball each time to load the barrel



Prototyping

Initial Prototype: Launching Barrel & Launch Plate

What we did:

We prioritized building the barrel and launch plate first so we could design the rest of the system around how this core assembly would be secured.



What we learned:

The barrel must be smoother to let the launch plate and ball pass freely.

What we changed:

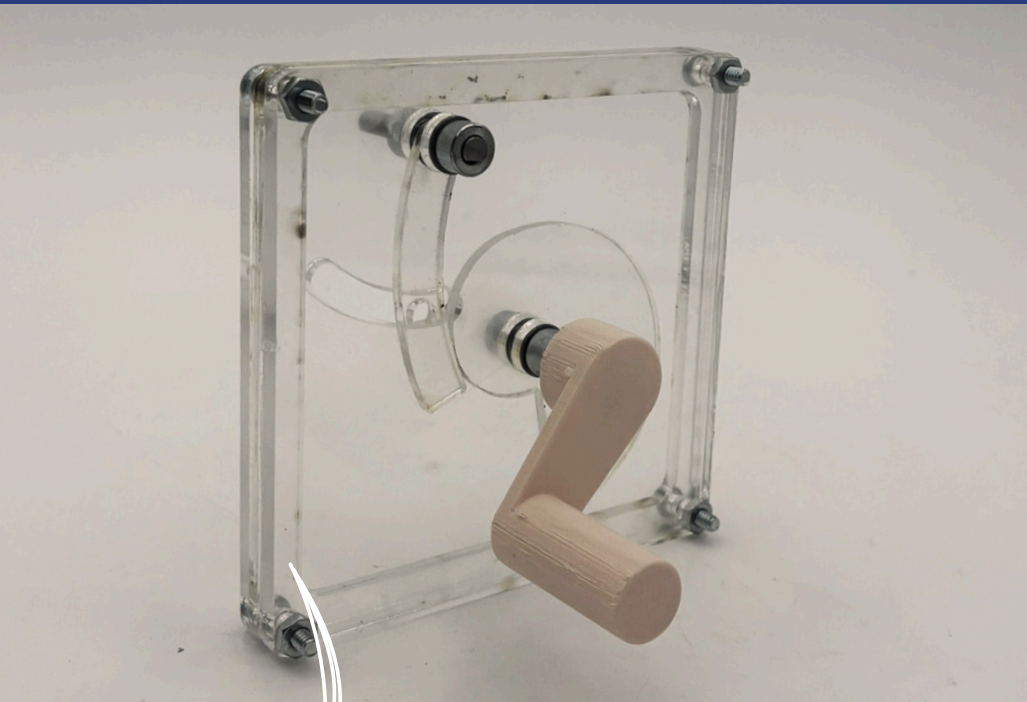
We used smaller print layers, enlarged the exit, and split the barrel into two pieces for easier sanding and launch plate insertion.



Critical Part: Launching Gear Mechanism

What we did:

We designed a gear system that uses rotational motion to both load and launch ping pong balls. Our goal was to make the launcher efficient, intuitive, and less prone to user error by allowing the user to simply turn the gear, reducing external factors.



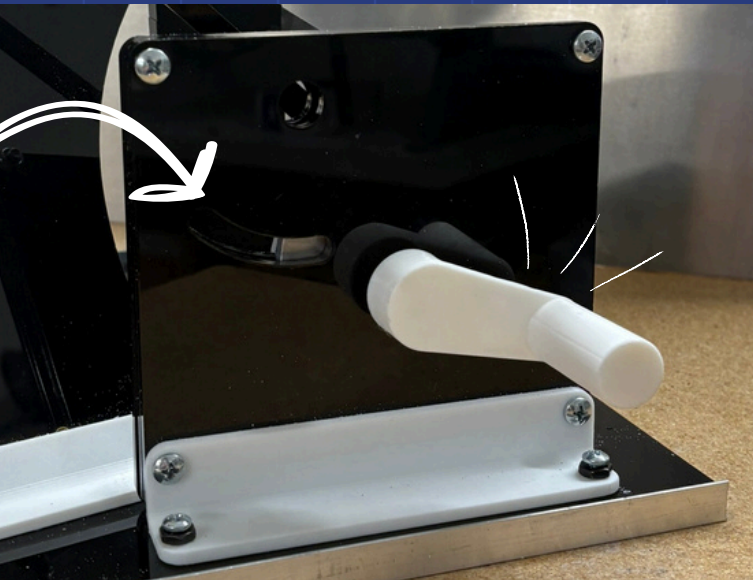
What we learned:

The gear needs to pull a very heavy load so it made it difficult to use the handle with all the friction from acrylic on acrylic and the shearing of the plastic.



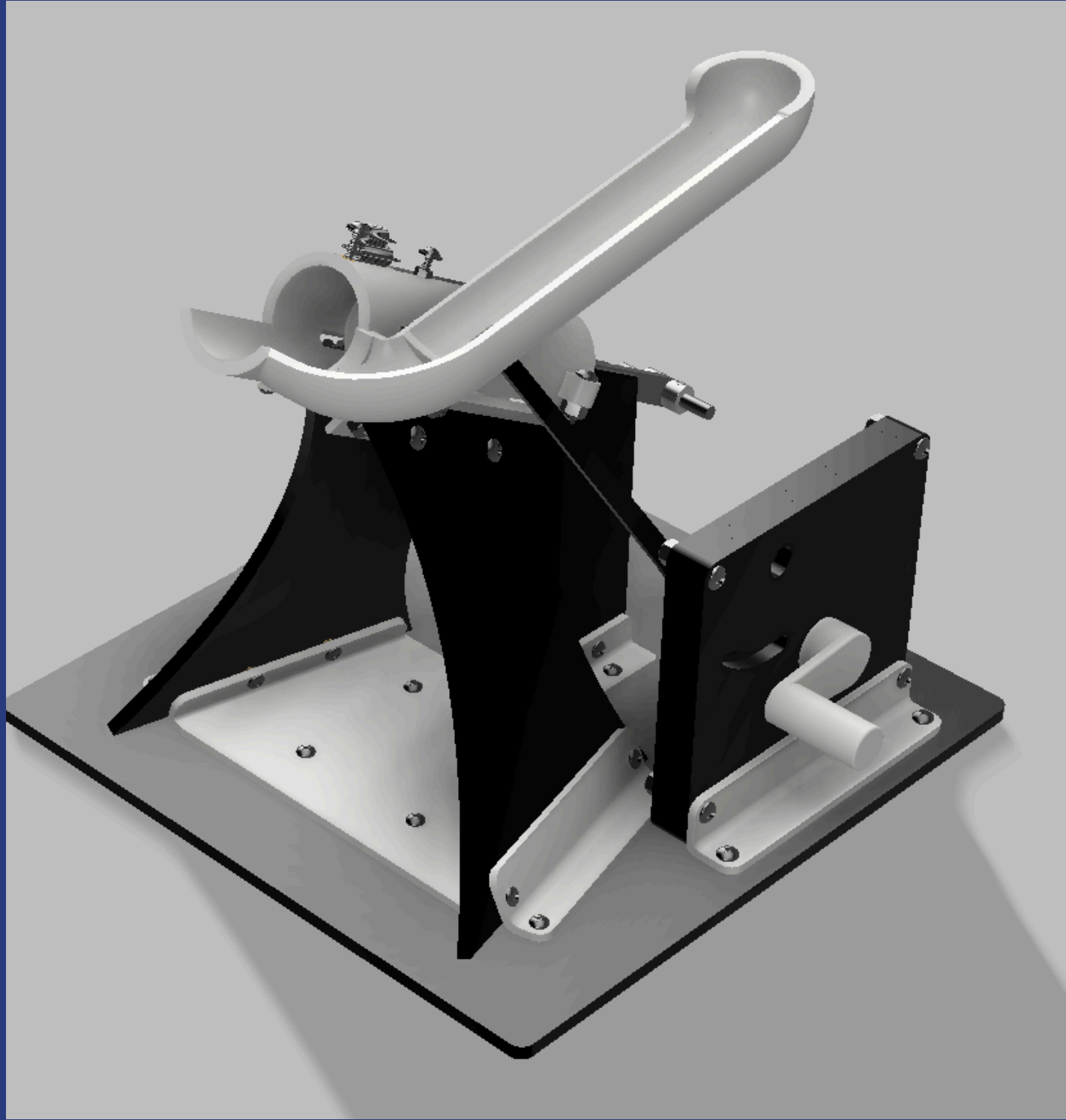
What we changed:

We adjusted the tolerancing of the gears and replaced parts of the acrylic parts with metal parts to reduce friction.

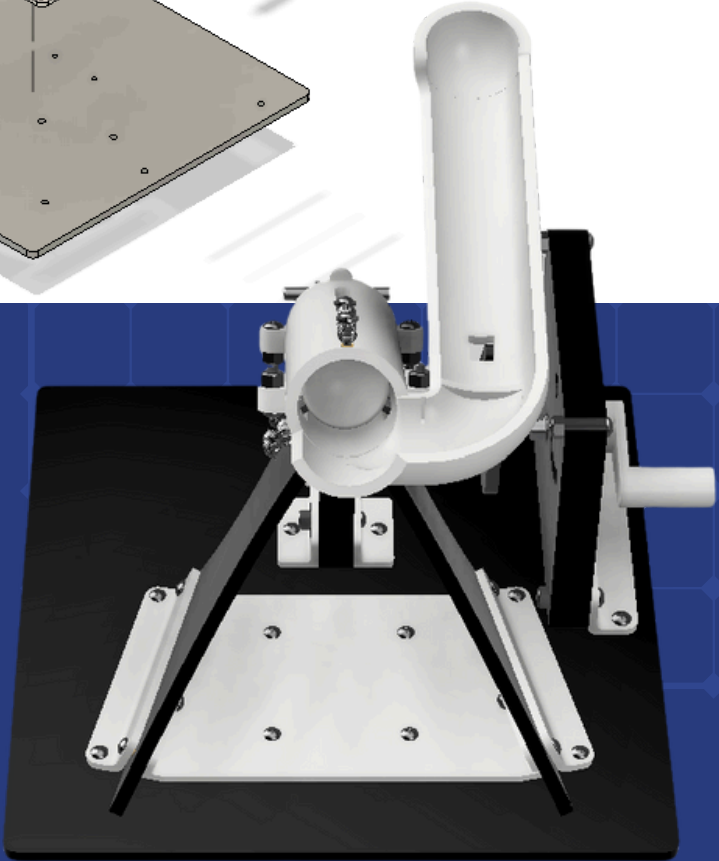
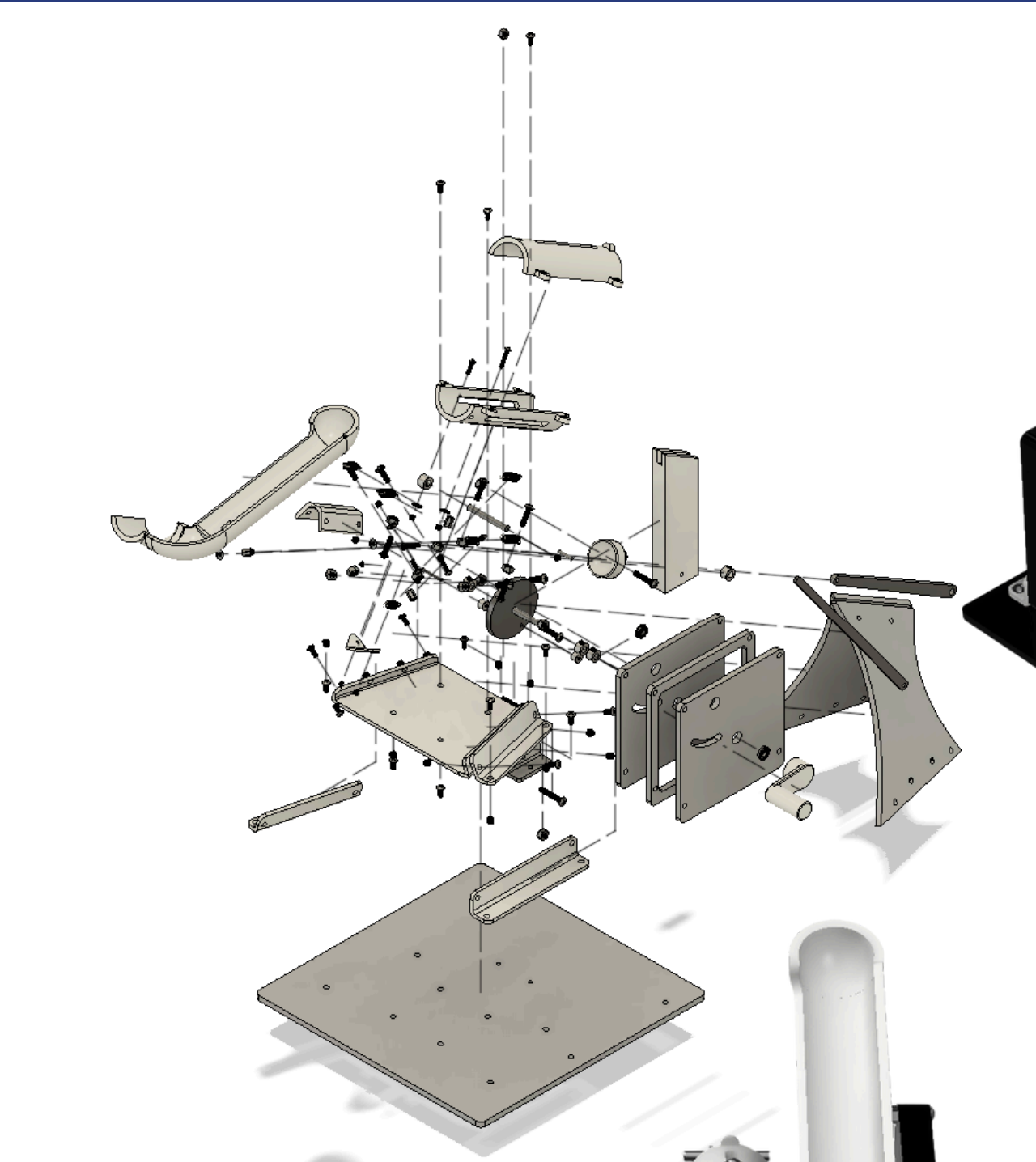


Redesign

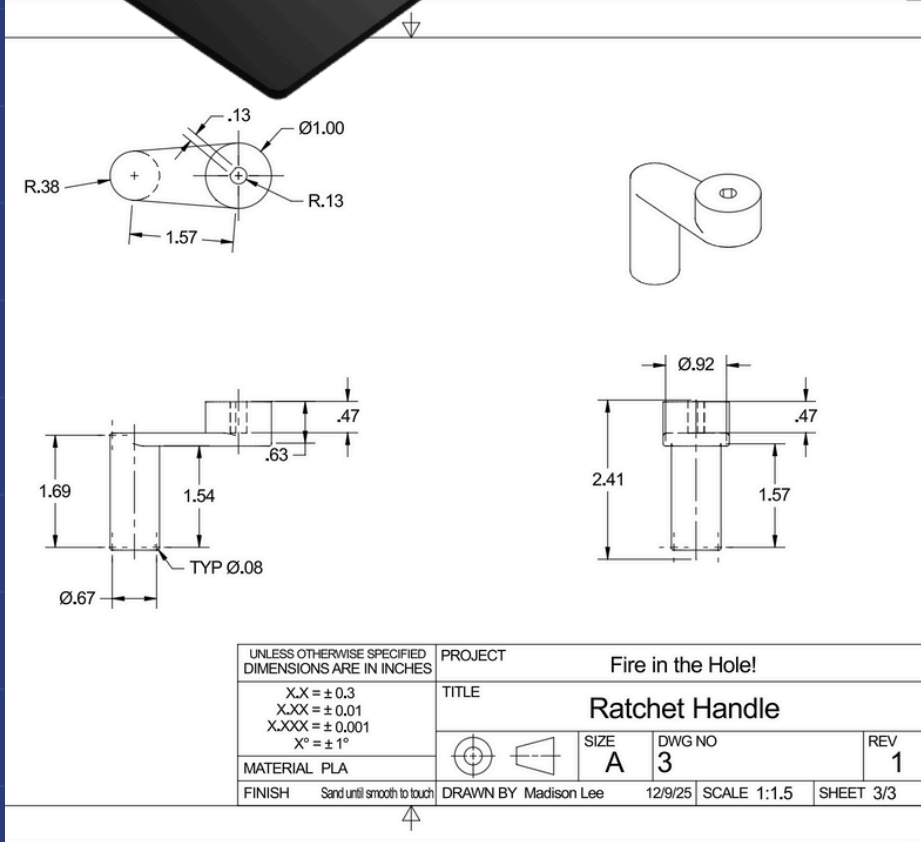
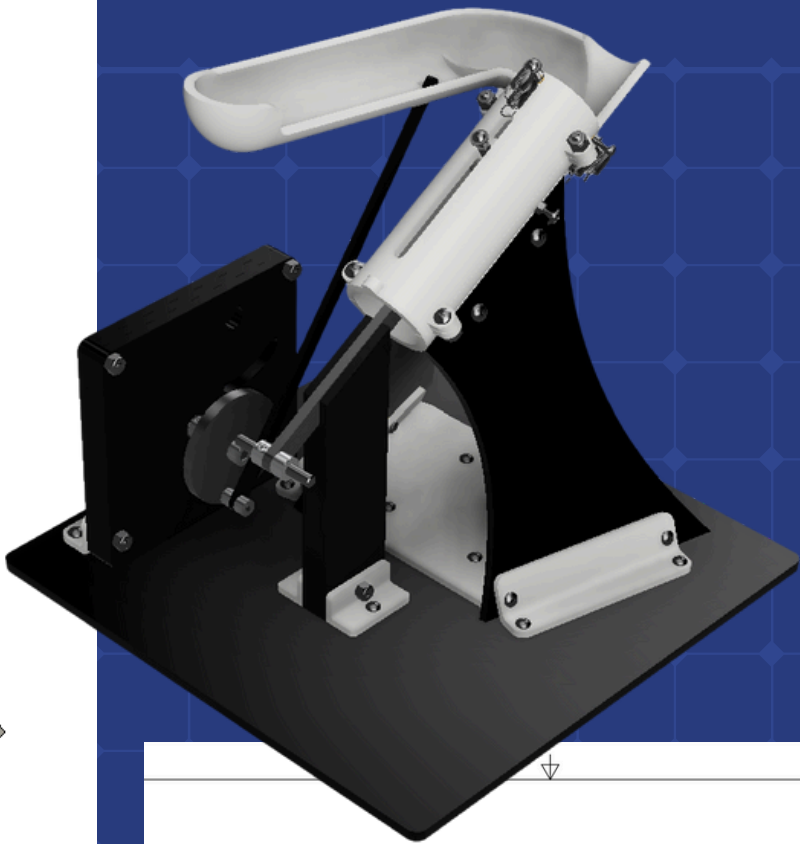
After testing, our main concern was whether the gear system could consistently generate enough force and travel to pull the lever arm far enough to launch the ping pong ball 8 ft.



We redesigned the launching mechanism to use a simple linear pull, while keeping rotational motion for the hopper mechanism.

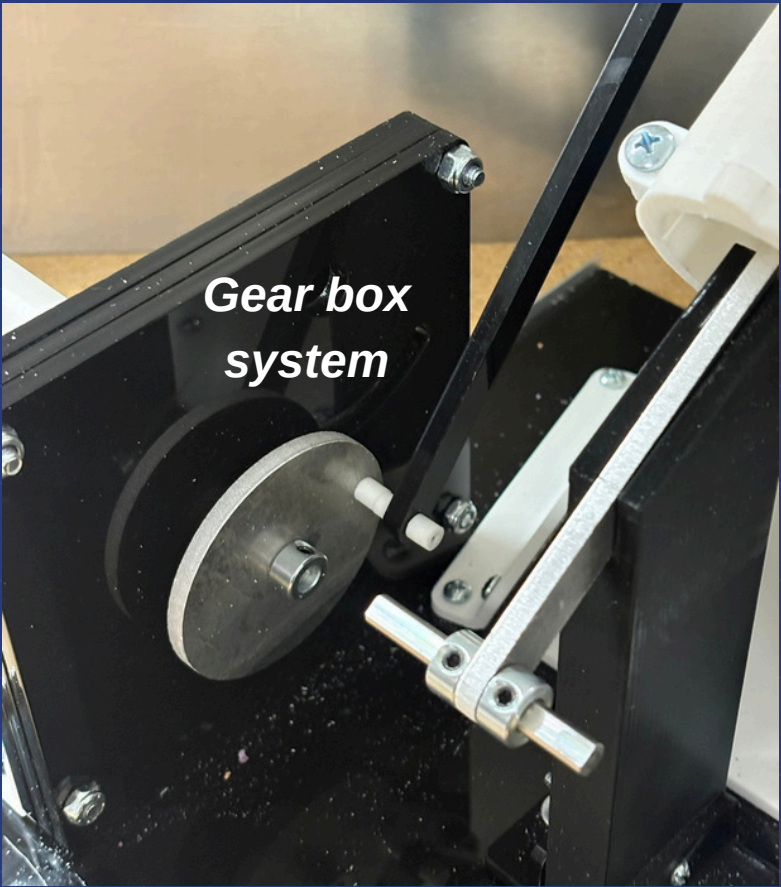
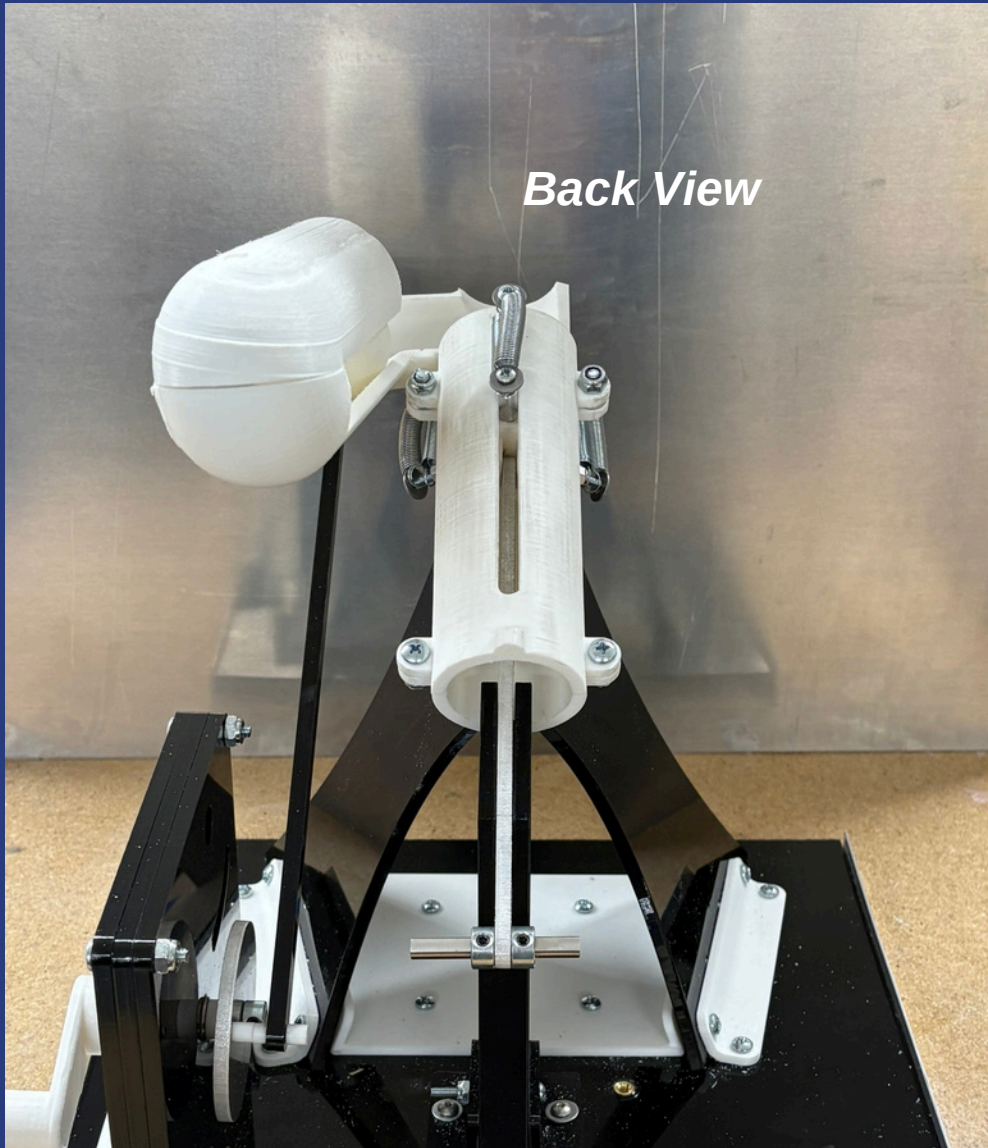
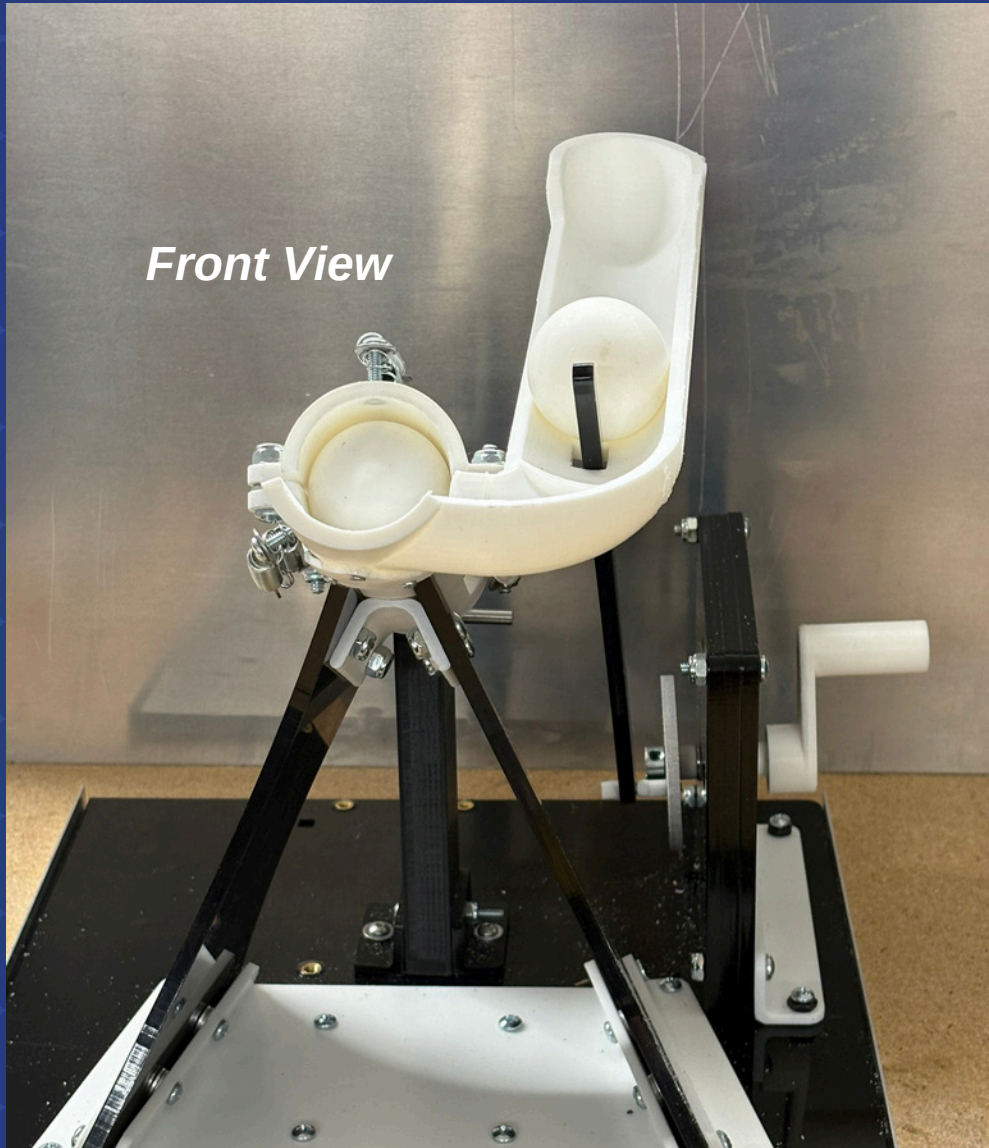


Exploded View of Assembly



Engineering drawing of the handle component

Final Prototype

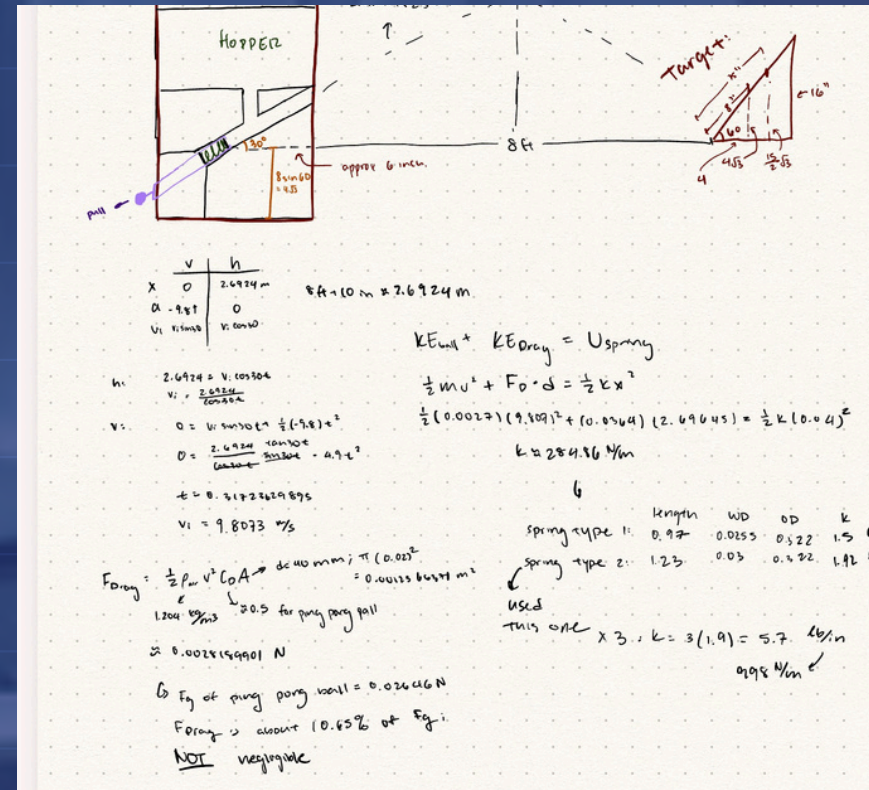


Analysis & Reflection

Analysis: Mathematical Approximations

Kinematics

We approached this project with our initial calculations in mind, however, these calculations severely underestimated the friction and air resistance that the ping pong ball faced so our initial prototype had a hard time launching as far as we hoped.



Reflection: Hindsight as Foresight

Challenges:

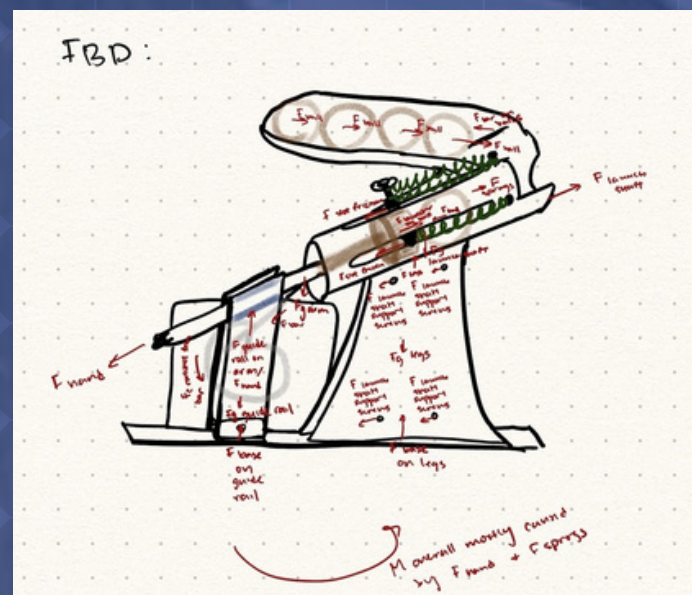
We took on a more ambitious and complex design. This pushed us to explore multiple mechanisms and try a more challenging system overall, even if it meant the ping pong launcher received less dedicated refinement. We faced challenges with this increased complexity and ultimately had to pivot with the limited time we had at the end.

Successes:

Once we realized certain aspects of our initial design wouldn't work, we quickly identified what could be kept, what needed improvement, and what required a full pivot. This streamlined our redesign process and allowed us to rebuild a launcher capable of sending a ping pong ball 8 ft (within just a few hours).

Future Improvements:

Our next step would be to redesign key components so the rotational-motion system functions as intended. This includes reinforcing load-bearing parts and improving alignment between the shaft, gears, and spring to ensure smoother motion and more consistent launches.



Force Analysis on Launcher

An analytical breakdown of all the forces acting on our system from the hopper system all the way down to the handle that the user grabs.